

31338 Network Servers

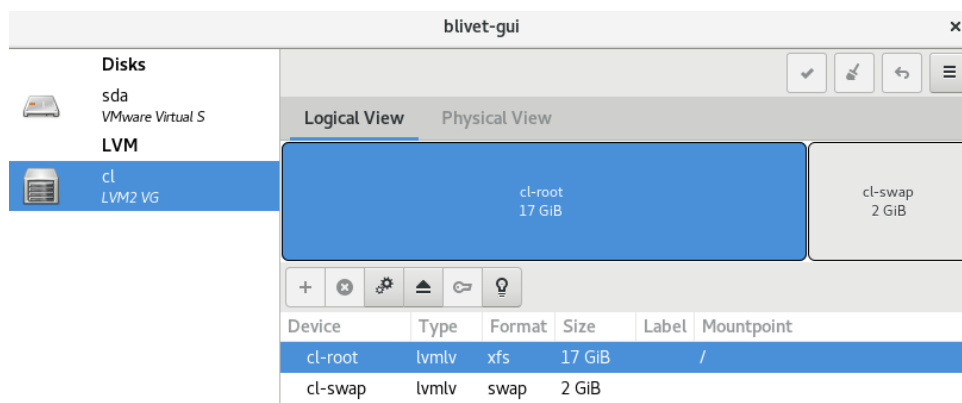
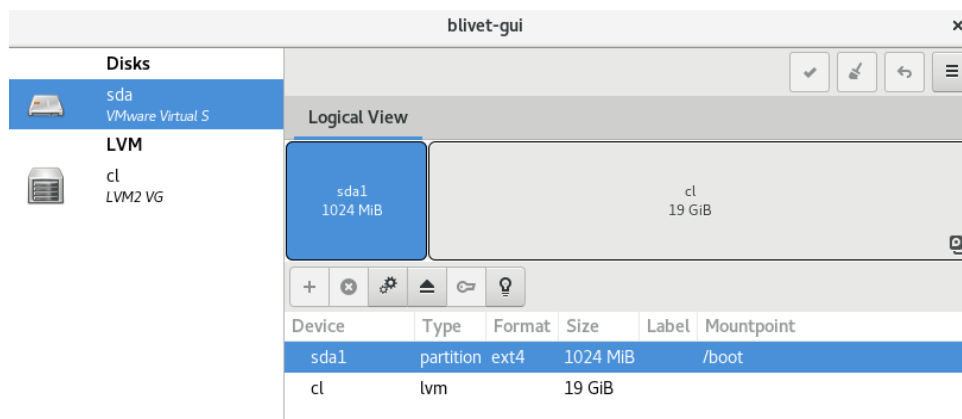
Lab 2d: Disk partitioning

Aims

1. To be able to partition your hard disk

Task 1: Linux disk partitioning

By default, Redhat and Centos derivative Linux operating systems first create a disk partition called /boot, just for the bootloader files. Then for the remaining disk space, they use the Logical Volume Manager to create 2 logical volume groups called cl-root and cl-swap.



This allows the system to expand the size of the root drive as needed by simply adding more hard drives to the physical volume called “cl” in the above diagram

We have chosen in our configuration not to do this. After all, a USB hard disk is hard to expand any further!

What do you notice about the boot drive? Is it managed by the Logical Volume Manager?

Why do you think it is done this way?

(hint: Think about how can you boot before the LVM starts, or booting into single-user mode?)

On our lab image we have a simple 2 partition layout based on the BIOS MBR default partition layout.

First thing would be to check the current layout: Use the following command

`mount`

There's a lot of output. Let's filter it using `grep` to only show the lines that start with `/dev`:

`mount | grep '^/dev/'`

This tells you the device (hint: `/dev/sda*`) and where it is mounted on the file system hierarchy. Run the command:

`df`

This tells you the amount of free space in the mounted file system. You can use the `-h` option to show MB/GB.

Finally, try:

`parted /dev/sda print`

This should display your current partitions. To see what's inside the logical volume, use:

`lvs`

To view more detail of the swap partition (like how much swap memory is currently being used), try:

`swapon`

Open two shell windows. In one of them, run the `top` command. Pay close attention to the swap space shown by `top` (the "MiB Swap" line). Run the following commands. Pause between each command and watch what happens.

`swapoff -a`
`swapon -a`

GUI interface

Use your skills in finding and locating software by installing the following software packages one after the other:

- `epel-release`
- `blivet-gui`

Run `blivet-gui` and you should see a graphical view of the disk layout.

Note that our disk is called the VMware virtual storage (VMware hard drive).

Task 2 (optional): Creating partitions

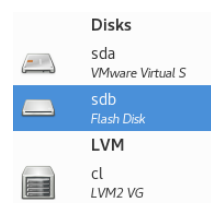
If you have a spare USB flash drive, even a small one, and you are prepared to delete all of its contents, continue with this task and we will create some new partitions. But be aware:

THIS WILL DESTROY ALL DATA ON YOUR SPARE USB FLASH DRIVE

Insert your spare USB flash drive into your computer, and if prompted by VMware whether you want to attach it to your host machine or the guest VM, choose to attach it to your Linux VM.

Reopen `blivet-gui`, and you should now see the additional drive appear, with the words "Flash Disk" underneath it. It will probably be the device named "sdb".

Choose the "sdb" flash drive disk (very important – make sure you pick the right one!). If it was mounted, it needs to be unmounted. Choose the icon that looks like "eject".



We want to create 2 partitions:

- A 1GB EXT4 partition called "opt". Format this drive.
- A 1GB swap space partition called "swap2"

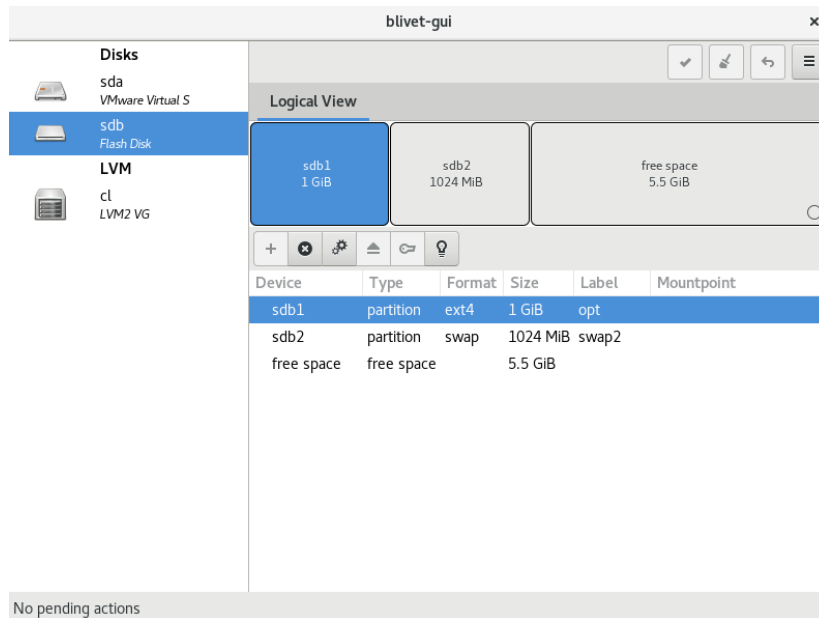
From the settings menu (looks like wheels/cogs), choose "Format". Change the partition format to "ext4" and use the label "opt". Then choose to resize it to 1GB. Click on the "free space" line and use the "+" to add a new partition. This time make it a swap partition, 1GB in size and with label "swap2".

Note that as you've been doing this, at the bottom-right of the blivet-gui window it shows "X pending actions". That's because it has not actually applied your changes yet, it has put them in a queue to be processed. You can view the pending changes by clicking on the "pending actions" link.

When you are satisfied with the changes you're about to make (and you are absolutely sure they will be applied only to your spare USB flash drive), then you can press the tick button near the top-right to apply all pending actions.



Your partitions on the USB flash drive should then look like this:



It shows:

- sdb1 is a 1GB ext4 partition, with label "opt"
- sdb2 is a 1GB (1024MB) swap partition with label "swap2"
- The remainder of the disk is free space (here we used an 8GB flash drive)
- At the bottom-left, it shows "No pending actions", i.e. all changes have been applied.

First let's use the swap. Run the following commands. If your flash drive is something other than "sdb2" then change it when you type the commands.

```
mkswap /dev/sdb2
swapon /dev/sdb2
swapon
```

What does the last line (swapon) display now? How is it different from when you ran it before? Run the **top** command again. What does it say about swap space now?

Now let's mount the new ext4 partition on the /opt directory (mount point).

First, create some file in the /opt directory. E.g

```
touch /opt/hello.txt
```

Verify it's there using the **ls** command.

Then mount the new partition onto the /opt directory with the following command (again if your flash drive partition is something other than "sdb1" then change it in the command).

```
mount /dev/sdb1 /opt
```

Now look for your hello.txt file again. Where is it?

Create a new file in /opt called "world.txt":

```
touch /opt/world.txt
```

Verify it's there with `ls`.

Unmount the partition with:

```
umount /dev/sdb1
```

(Note: the command really is “umount”, not “unmount”).

Use ‘ls’ to look at what’s in the /opt directory now. What’s going on? Try mounting and unmounting a few times and check the /opt directory. Note your findings in your journal.

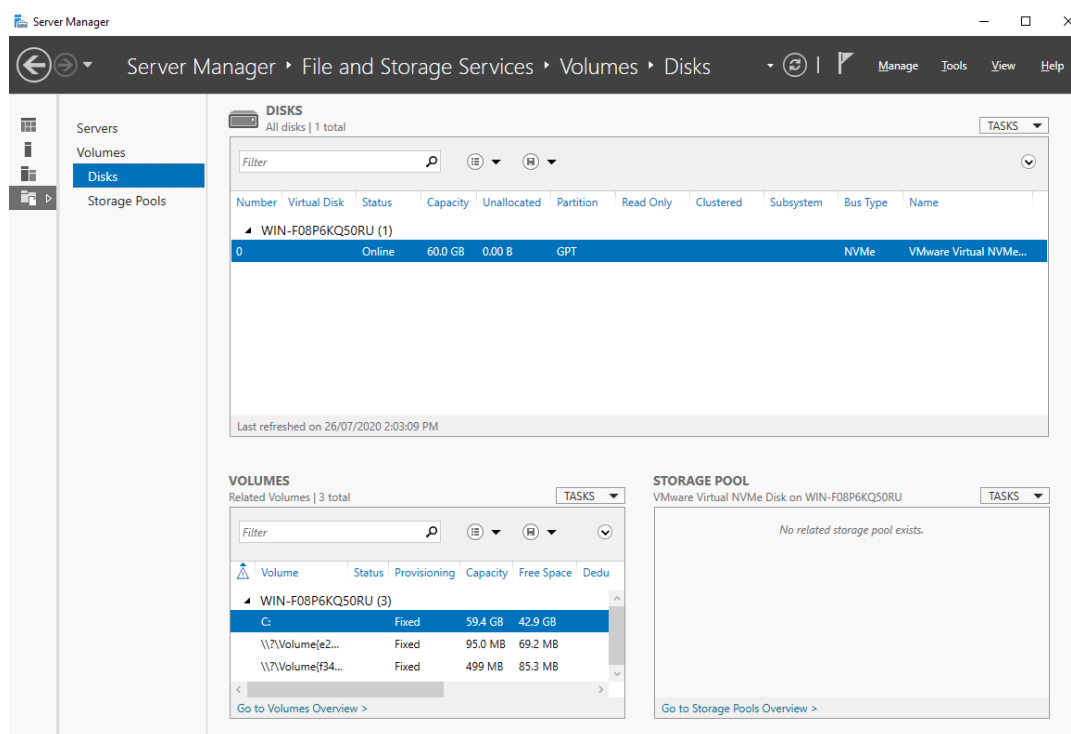
Finally, to end the activity, before you remove your USB flash drive, run:

```
swapoff /dev/sdb2  
umount /dev/sdb1
```

If it tells you the device is in use, you probably have a shell window open that is in the /opt folder. When everything is unmounted, you can remove the USB flash drive.

Task 3: Windows Server Disk Management

Server Manager on Windows has a separate dashboard for managing ‘File and Storage Services’. Find it in the left-hand menu of Server Manager after your Windows Server boots. You should see four options: Servers, Volumes, Disks and Storage Pools. Start by choosing “Disks” and your window should look something like the following.



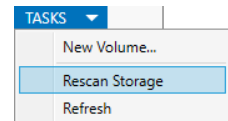
It shows that our system has a 60GB disk, with three partitions/volumes. The main volume is the one labelled “C:” which is the main OS hard drive.

Have a look around the way you did in Linux, and make notes in your journal about the partition structure.

If you used a spare USB flash drive to create partitions in Linux, let’s mount it here in Windows too (optional if you skipped that part in Linux because you didn’t have a spare flash drive).

Insert the USB drive into your computer. When prompted by VMware whether to connect it to the host or guest OS, connect it to the Windows Server virtual machine.

You'll probably notice that in the File and Storage Services window, it doesn't automatically appear. That's OK. Near the top-right of the window you should find a menu labelled "TASKS". From the tasks menu you can "Rescan Storage" and your USB flash drive should appear.



Your USB drive should now appear in the list of disks.

Number	Virtual Disk	Status	Capacity	Unallocated	Partition	Read Only	Clustered	Subsystem	Bus Type	Name
WIN-F08P6KQ50RU (2)										
0		Online	60.0 GB	0.00 B	GPT				NVMe	VMware Virtual NVMe...
1		Online	7.50 GB	5.50 GB	MBR				USB	Generic Flash Disk

Note that if it's the same disk you used earlier, you can still see that some of the space is used. In our example, we used an 8GB flash drive (Windows shows capacity as 7.5GB), but it shows only 5.5GB unallocated. This is because we created 2 x 1GB partitions in Linux. Although Windows can't read the filesystem types, it knows the partitions exist. How?

Let's clear everything on this drive and start again. Right-click on the USB flash drive (choose carefully!) and select the "Reset disk" option. This will clear all existing partitions. How can you tell it worked?

Now right-click on the same drive again and choose "New Volume...". A wizard appears. Follow it through and create a volume (partition) with drive letter "N" that uses all available space on the disk. Note that you could also assign it to a folder, which is a bit like a mount point in Linux.

Verify that your new N: drive exists, and its size.

The good news is that now your flash drive is back to being a normal Windows flash drive that you can use in other computers!

Managing disks on non-server Windows systems

The tool we have been using – File and Storage Services within Server Manager – is specific to Windows Server machines. It isn't normally available on Windows desktops.

To manage disks and volumes on a Windows desktop machine, go into the Settings and search for "disk" and you should find the Windows disk management tool ("Create and format hard disk partitions"). You can do the same thing on your Windows Server.

